EmotiSync: Accuracy & Performance Benchmark

Empirical Neural Network Evaluation on 921 Labeled Facial Images

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Audience: Executive Technical Review
Repository: github.com/ANKARAHAMSA/emoti-sync
Pipeline: TinyFaceDetector + FaceLandmark68 + FaceExpressionNet

ACCURACY ON DETECTED FACES
60.5%
95% CI: 56.8% – 64.1%

COARSE POLARITY ACCURACY
71.5%
Positive vs Neutral vs Negative

HAPPY CLASSIFICATION (F1)
87.7%
Precision: 84.9% | Recall: 90.7%

RANDOM GUESSING BASELINE
14.3%
1 in 7 uniform baseline

1. Executive Summary & Test Methodology

This evaluation measures the real accuracy of **EmotiSync** on a class-balanced test set of **921 real-world labeled facial images** (up to 150 images per emotion category). Testing was conducted using the identical deep learning models bundled in the app (`@vladmandic/face-api` 1.7.15) running in an isolated headless WASM environment to ensure absolute parity with in-browser execution.

Core Verdict: The model achieves **60.5% overall accuracy** across all 7 emotions (over 4× better than random guessing at 14.3%), and **71.5% accuracy** when grouping emotions into Positive, Neutral, and Negative sentiments. **Happy** (87.7% F1) and **Angry** (70.3% F1) are highly reliable, while **Fearful** and **Disgusted** show lower accuracy and are frequently categorized as Neutral.

2. Emotion-by-Emotion Accuracy Breakdown

Emotion Category	Support (Total)	Detected Faces	Precision	Recall	F1 Score	Reliability Rating
Happy	150	118	84.9%	90.7%	**87.7%**	High (Production Ready)
Angry	150	123	73.5%	67.5%	**70.3%**	Good (Consistent)
Surprised	150	120	73.9%	54.2%	**62.5%**	Moderate
Sad	150	104	56.1%	57.7%	**56.9%**	Moderate (Confused with Neutral)
Neutral	150	122	42.4%	68.9%	**52.5%**	Over-represented (Model Default)
Fearful	21	16	12.5%	12.5%	**12.5%**	Low (Limited Sample)
Disgusted	150	68	21.7%	7.4%	**11.0%**	Low (Misclassified as Neutral)

3. Confusion Matrix (Error Distribution)

Rows represent the **True Emotion**; Columns represent the **Model's Prediction**. Diagonal entries (blue) represent correct predictions.

True Label \ Predicted	Happy	Neutral	Surprised	Sad	Angry	Fearful	Disgusted
Happy	107	6	1	3	0	1	0

True Label \ Predicted	Happy	Neutral	Surprised	Sad	Angry	Fearful	Disgusted
Neutral	5	**84**	6	12	9	3	3
Surprised	1	23	**65**	10	13	6	2
Sad	5	32	1	**60**	2	1	3
Angry	2	14	8	6	**83**	1	9
Fearful	0	1	5	5	2	**2**	1
Disgusted	6	38	2	11	4	2	**5**

4. Confidence Calibration & Reliability Thresholds

The model's raw probability output was analyzed against actual correctness. When the model outputs high confidence (≥90%), its empirical accuracy rises to **70.8%**. Conversely, when confidence falls below 70%, accuracy drops to only **37.6%**.

Production Guardrail: In the updated application, predictions with confidence under 70% are tagged with an *"Unsure"* status indicator to maintain user transparency.

Reported Confidence	Share of Total	Real-World Accuracy
Below 50%	~4%	35.7%
50% – 70%	~16%	37.6%
70% – 90%	~21%	54.3%
90% and Above	~59%	**70.8%**

5. Findings & Strategic Recommendations

Key Takeaways for Stakeholder Presentation:

- **Effective for Sentiment & High-Confidence Signals:** The system is highly dependable for Positive/Happy (87.7% F1) and Angry/Tension (70.3% F1) detection, achieving 71.5% sentiment polarity accuracy overall.
- **Temporal Multi-Frame Smoothing:** Unlike static single-image tests, live webcam analysis utilizes EmotiSync's cross-frame tracking (IoU FaceTracker), preventing momentary flicker and providing a significantly smoother real-time user experience.
- **100% Privacy & In-Browser Execution:** Zero facial images or telemetry data ever leave the client device; all inference executes entirely on-device via WebAssembly/WebGL with bundled weights.
- **Recommended Usage:** Ideal for live user sentiment, attention engagement, and mood trend monitoring over time. For strict 7-class forensic classification, confidence filtering (≥70%) should be applied.